Elucidate and Object Detection as a Service (ODaaS)

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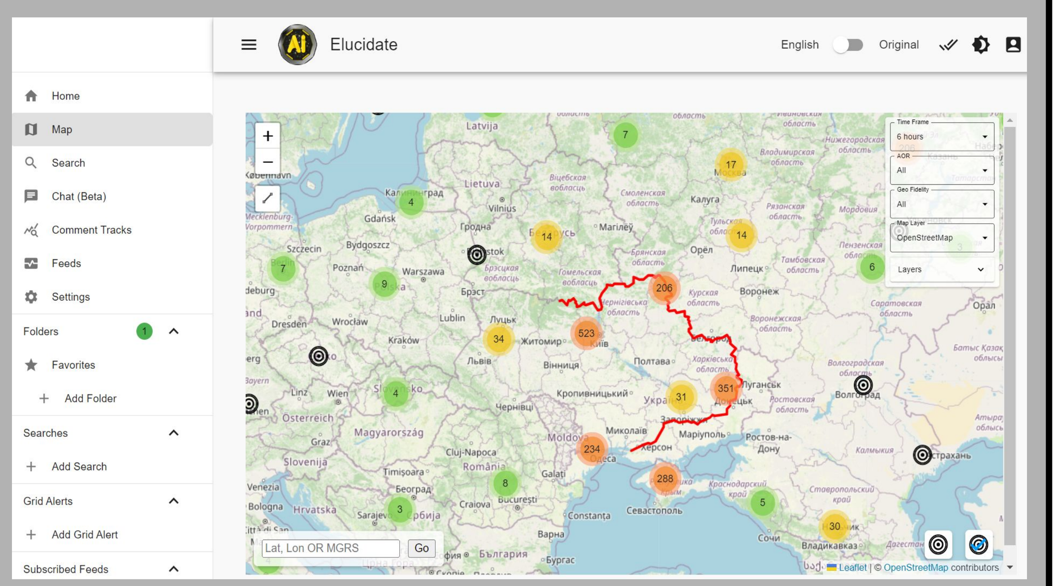
Elucidate Overview



- Provides a platform for Soldiers to gain information advantage and broad situational awareness through publicly available information.
- Use AI tools and readily constructed filters/searches in order to organize data and shorten the length of time it takes for analysts to find actionable information. (NER, Object Detection, Recommender Systems, Translation, RAG/LLM)
- Provide workflow optimizations for processing information rapidly and sharing with partners in a unified platform.

Problem Definition

- The current iteration of Elucidate is not capable of recognizing images or videos that are not associated with text in a social media post.
- There is currently no established environment with which to deliver a Computer Vision Object Detection Model (CV-OD).

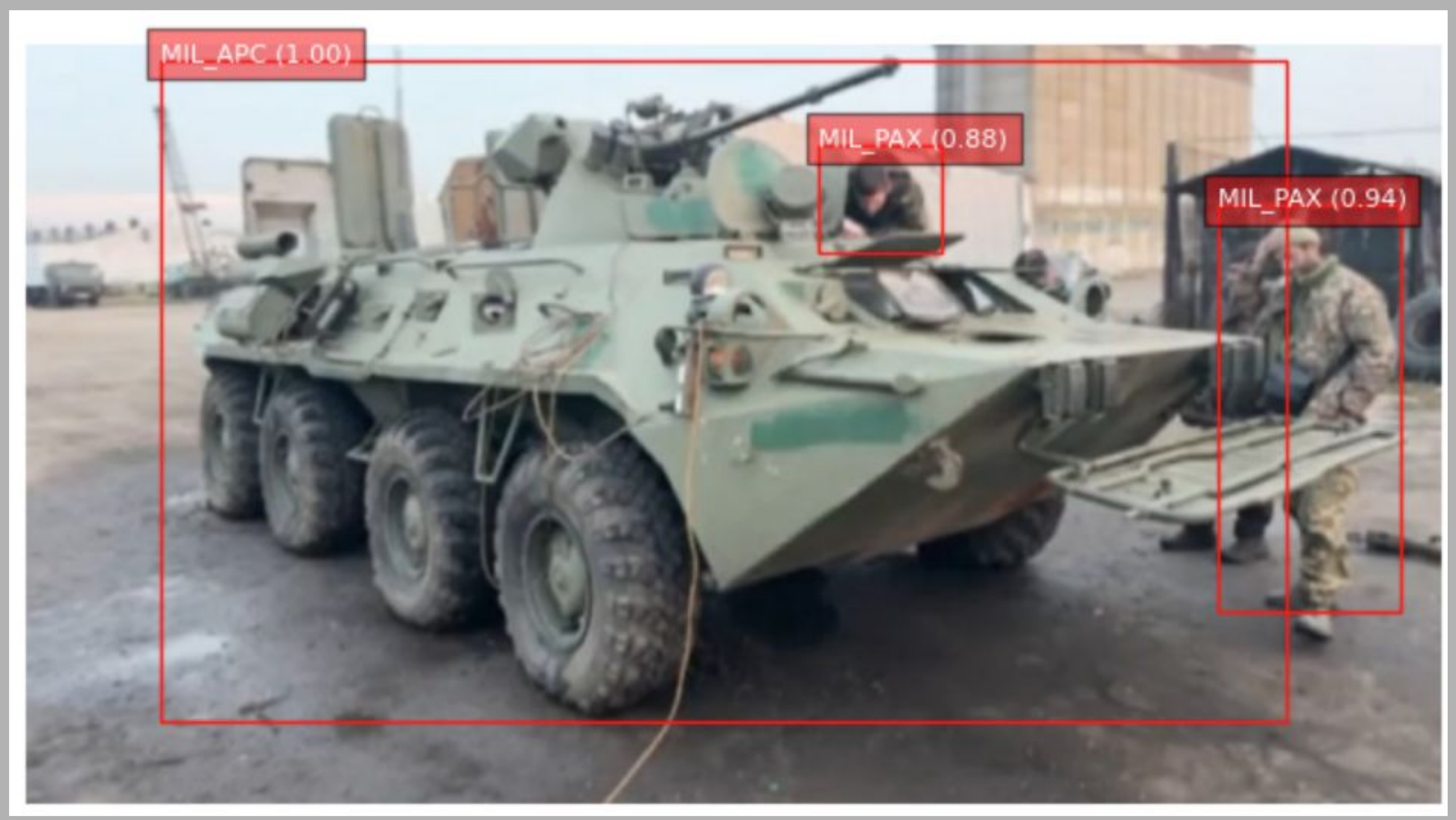
Project Overview

Goal	Provide object detection service to Elucidate by developing a GOTS (Government off the Shelf) solution to by taking a platform agnostic approach to make the application as broadly accessible as possible across DoD and partner forces.
Audience	Security Assistant Group - Ukraine (SAG-U) analysts using Elucidate for daily analytic workflow
Scope	Model that can recognize military vehicles and personnel in unlabeled social media photos
Key features	Object detection, Platform agnostic, GOTS solution, Scalable

Current Solution

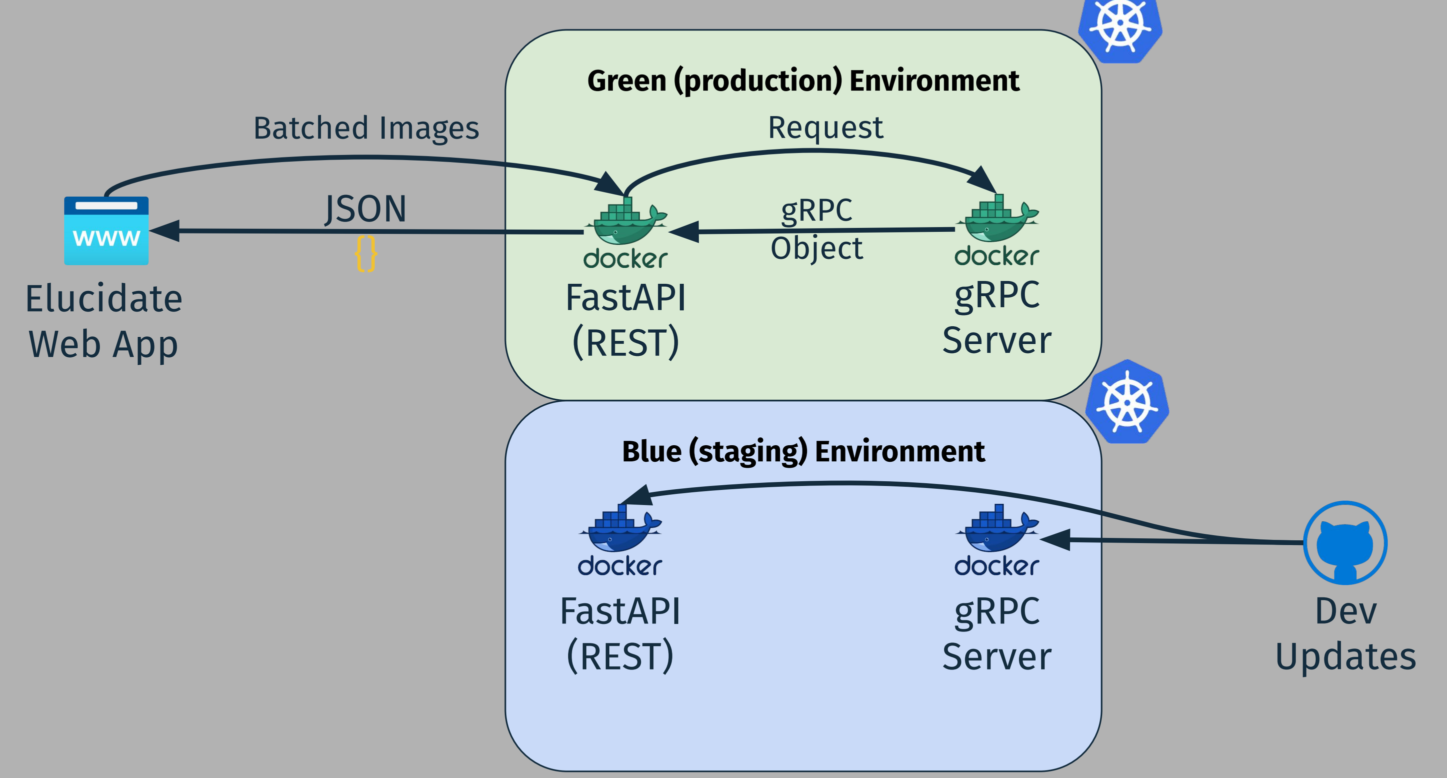
Service Integration with Elucidate

The Elucidate object detection model is engineered to autonomously identify and categorize images featuring military vehicles and personnel. It enables coarse-grained indexing of unlabeled images, ensuring they surface during keyword searches conducted by analysts. With this enhancement, the Elucidate application can overcome traditional text-based limitations, offering a more comprehensive and intuitive analysis tool.



Service Architecture

Elucidate is an external application connecting to our microservices.



Ontology

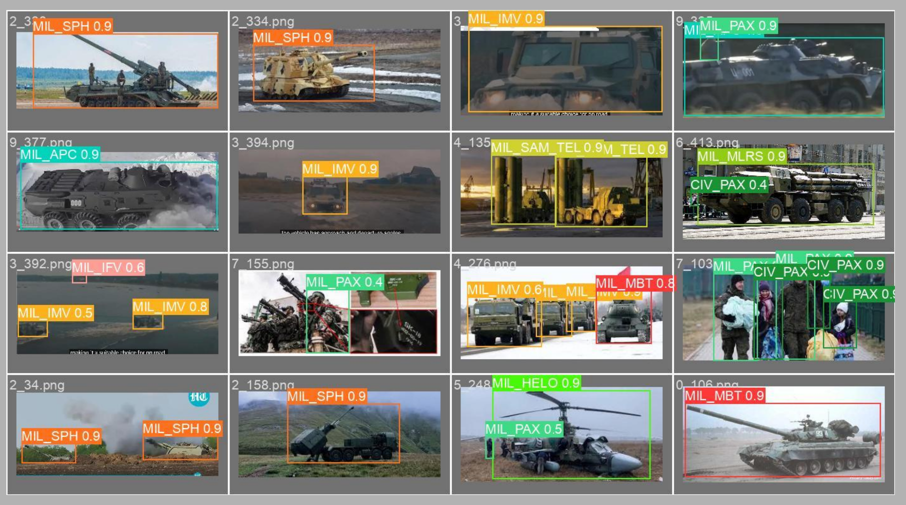
Classification	Description
MIL_HELO	Helicopters
MIL_MBT	Main Battle Tanks
MIL_APC	Armored Personnel Carriers
MIL_IMV	Infantry Mobility Vehicles
MIL_SPH	Self-propelled Howitzers
MIL_SAM_TEL	S-300 Surface-to-air-missile systems
MIL_MLRS	Multi Launch Rocket Systems
CIV_PAX	Personnel not in military uniforms
MIL_PAX	Personnel wearing military uniforms
MIL_IFV	Infantry Fighting Vehicles

The ontology underpinning the object detection model is tailored to the operational requirements of the end-users, acknowledging that Elucidate operates as a human-in-the-loop system. With this premise, we've adopted a broad granularity approach to the classification schema. This strategy categorizes military vehicles primarily by their functional roles, allowing for a streamlined initial sorting process. Subsequent, more detailed classifications are delegated to the expertise of intelligence analysts, who can apply their specialized knowledge, as necessary.

Experimental Setup and Analysis

Labels vs. Predictions

1. **Labels (Ground Truth):** These images show the **actual objects** that were manually annotated before running the model.
2. **Pred (Predictions):** These images display the objects **predicted by the YOLO model**.



Results

- Each set (Labels and Pred) contains images showing various objects such as Self-Propelled Howitzers (MIL_SPH), Infantry Mobility Vehicles (MIL_IMV), and Armored Personnel Carriers (MIL_APC).
- Comparing the "labels" images with the "pred" images helps us understand the model's accuracy.
- If the boxes and labels in "pred" match those in "labels," the model made correct predictions.

Conclusion

- By comparing the "labels" and "pred" sets, we can assess the YOLO model's performance.

Results and Key Takeaways

Blue/Green Deployment Method

Allows for **continuous updates** without service interruptions.

Modular Service

Allows for updating only **relevant code** without the need to touch other services

Model updates can be pushed to the gRPC server without the need to change the FastAPI server and vice versa.

More Data is Not Always Better

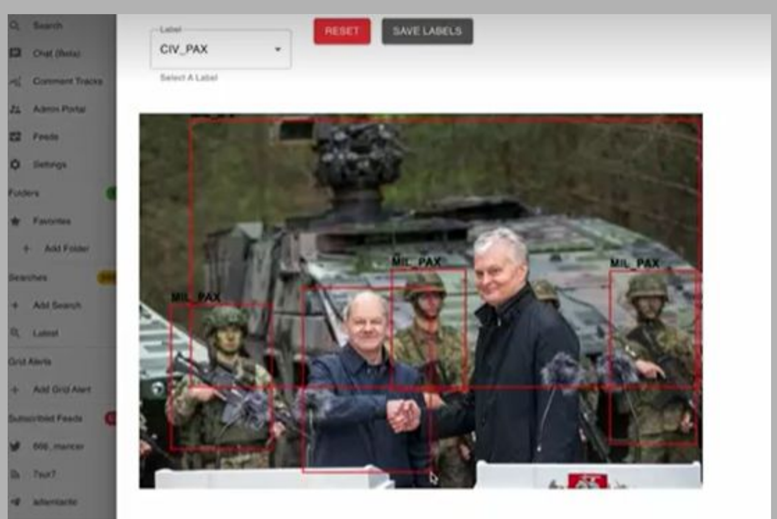
While we were fortunate to receive a large amount of labeled data through a partnership with CDAO, the imbalance of images per class made it largely unusable for re-training our model.

Future Work

Currently in production for a future release of Elucidate, analysts will have the ability to tag images and relabel images in the user interface.

Data Will be collected continuously with user feedback. A thumbs up and down button will be implemented for users on each image.

This will retrain the Object Detection Model and allow to provide better predictions.



AI Technicians 2023-2024 - Capstone Project

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